

Human-AI Chat and Inbox Collaboration Scenarios Research

Document ID: RES-017 **Last updated:** 2026-07-16 **Status:** Research / design — no runtime implementation in this loop **Trigger:** Owner feedback requesting a follow-up to RES-015 and RES-016 to ensure that:

- When AI agents are conversing with each other, the owner can at any time join/intervene in the chat to continue or asynchronously guide the topic.
- The different scenarios of owner-AI collaboration via the Chat Room (聊天室 / AI 對話) and the Inbox (收件區) are systematically organized and reconciled.

1. Source Basis

Local sources reviewed:

- docs/07_research-and-design/RES-015_ai-chat-reference-context-and-cross-model-collaboration-research.md — central in-context reference context design and A2A consent/deadlock delegation.
- docs/07_research-and-design/RES-016_module-scoped-file-and-media-library-tab-and-classification-routing-research.md — module-scoped files/media library tabs and classification rules.
- docs/07_research-and-design/RES-014_ai-chat-thread-organization-rename-autotitle-and-multi-agent-display-research.md — chat thread folders and linking AgentBusTask multi-agent threads read-only in the sidebar.
- docs/07_research-and-design/RES-010_cross-module-human-ai-agent-operating-model-research.md & RES-011_human-ai-event-operating-model-research.md — define the Inbox discussion Thread and InboxItem contracts, the cross-agent invitation sequence (§13.1), and rule-memory/rejection-learning.
- docs/02_architecture-and-rules/ARC-032_internal-multi-agent-task-message-bus-contract.md — the multi-agent task and message bus contract.

2. Interaction Surface Map: Chat Room vs. Inbox

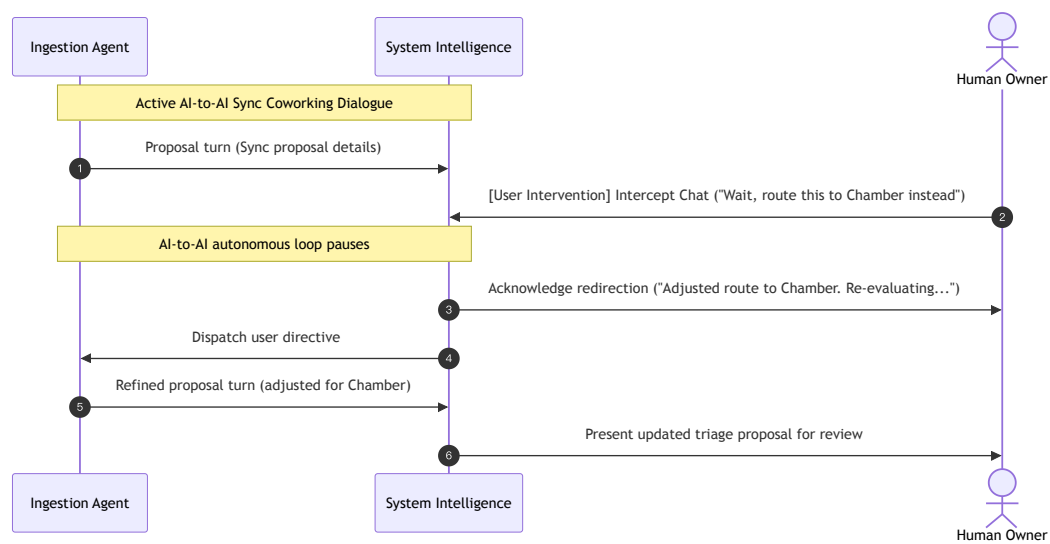
The system separates casual/creative discussion from formal/governed triage. The table below maps how these two workspaces function and cross-link:

Dimension	Chat Room (AI 對話 / /ai-input)	Inbox (收件區 / /inbox)
Primary Intent	Direct steering, exploration, casual brainstorming, multi-agent dialog intervention, and data referencing.	Structured judgment, approval of side effects, incident recovery, and consent dispute arbitration.
Data Nature	Free-form transcripts (ChatMessage list) with thread-specific reference context references.	Structured AnalysisEvent envelopes, markdown logs, version diffs, Action Plans, and question lists.
Agent State	Dynamic execution; agents run interactively in the composer loop.	Paused/Blocked execution waiting for explicit human authorization (waiting_approval).
Human Action	Send message, change active references, add folders, inline rename.	Approve plan, reject plan, answer question, write dispute resolution reason (Rule Memory).
Linkage	Can launch a detailed discussion thread linked to an Inbox Item (Scenario 3).	Receives automated escalations from Chat room consent deadlocks (Scenario 4).

3. Human-AI Collaboration Scenarios

Scenario 1: Synchronous User Intervention in AI-to-AI Chat

Definition: The owner opens an active multi-agent conversation thread in the sidebar (e.g., an ingestion sync coworking dialogue between a Sync Agent and System Intelligence) and types a message. The agents immediately yield to user intervention.

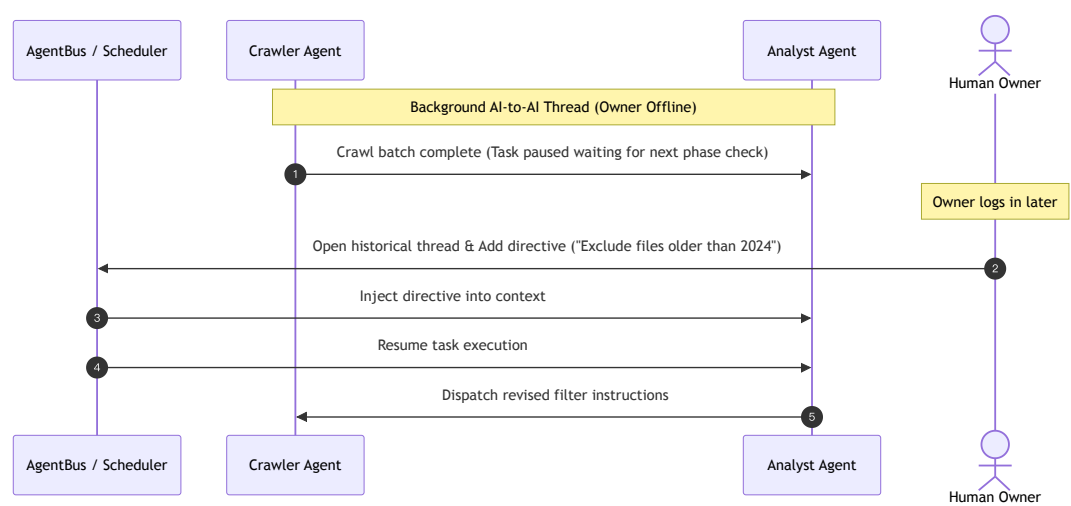


Key Constraints:

- **State Transition:** Typing a message as `authorKind: human` in a thread with multiple agents immediately pauses the autonomous agent turn scheduler.
- **Priority:** Human input is treated as a high-priority system prompt injection, overriding prior task instructions.

Scenario 2: Asynchronous User Intervention in AI-to-AI Chat

Definition: A background multi-agent task (such as a long-running research crawl or an overnight code audit) is running. The owner is not online. The owner later reviews the thread history, adds guidance at a historical point, and restarts the flow.

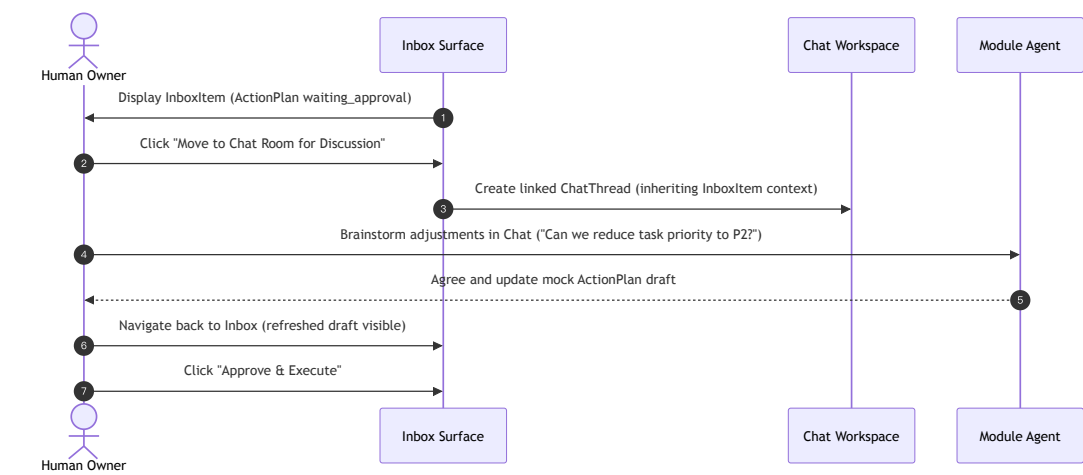


Key Constraints:

- **Checkpointing:** Long-running multi-agent tasks must save intermediate states (**checkpoints**) at the end of each logical phase.
- **Resume with Context:** When the owner inserts a message in a paused background thread, the task bus re-evaluates the context vector and updates active directives before prompting the analyst agents to resume.

Scenario 3: Discussing Inbox Items in the Chat Room

Definition: The owner receives a complex **AnalysisEvent** report in the Inbox (收件區). Instead of immediately clicking Approve/Reject, the owner wants to discuss the details with the module agent. The owner moves the topic to the Chat Room, aligns, and then returns to the Inbox to authorize.

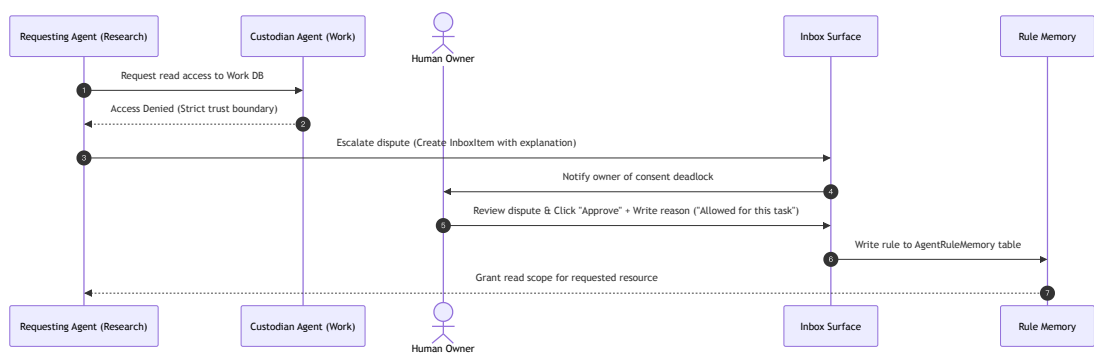


Key Constraints:

- **Linkage:** The created **ChatThread** stores a **LinkedInboxItemId** reference.
- **Draft Synced:** Any adjustments agreed upon in the chat conversation are saved in the temporary Action Plan draft associated with the origin **InboxItem**.

Scenario 4: Inbox Dispute Resolution for Cross-Agent Consent

Definition: Agent A wants to read Agent B's data but encounters a boundary dispute. They cannot agree, creating a deadlock. Agent A writes a dispute letter to the owner's Inbox. The owner decides and writes a reason, which is compiled into the Rule Memory.



Key Constraints:

- **Inbox Form:** The Inbox dispute triage item must render: Requesting Agent, Custodian Agent, requested resource scope, purpose description, and a text input for the owner to write the decision reasoning.
- **Rule Compilation:** The reasoning text is processed to append a structured row to the **AgentRuleMemory** table.

4. Backlog Rows (Updated under Phase 10)

Task id	Title	Module	Owner agent	Status	Priority	Risk	Dependencies	Files likely
AICHAT-010	Implement synchronous human intervention in chat loops	AI Input / Workflow	UIUXAgent, WorkflowAgent	TODO	P2	HIGH	AICHAT-004	src/app/input/ai task sched
AICHAT-011	Implement asynchronous guidance resume for background tasks	Agent Team OS / Workflow	WorkflowAgent	TODO	P2	HIGH	AICHAT-010	/src/lib command-
AICHAT-012	Implement Inbox-to-Chat discussion flow and draft synchronization	Inbox / AI Input	UIUXAgent, WorkflowAgent	TODO	P2	MEDIUM	AICHAT-006, EVENTOPS-007	/src/app/ai-inpu
AICHAT-013	Add acceptance and QA gate for Reference Context and Collaboration Scenarios	Acceptance / QA	QAAgent, ProductManagerAgent	TODO	P1	LOW	AICHAT-005	docs/08_qa/ACC-0 acceptan

5. NANDA Agent Protocol Gate

This research conforms to NANDA alignment rules (ARC-028):

- **Human Accountability:** The human owner is the ultimate arbiter of all multi-agent deadlock states.
- **Intervention Obs:** The agent scheduler must support pause and resume states that are observable in the append-only event log.
- **Dynamic Policy:** Consent rules are dynamically updated via AgentRuleMemory without requiring software restarts or code deployment.